ECON 003

Midterm Exam Study Guide

Saturday, August 15, 2026 | 10:00 AM – 12:00 PM (Pacific)
50 multiple-choice questions, drawn at random | Chapters 1, 3, 4, 5, 6

Key Notation

- **P** — price
- **Q** — quantity
- **QD** — quantity demanded
- **QS** — quantity supplied
- **D, S** — demand curve, supply curve
- **WTP** — willingness to pay
- **WTA** — willingness to accept
- **MB** — marginal benefit
- **MC** — marginal cost
- **MR** — marginal revenue
- **MP** — marginal product
- **TB, TC** — total benefit, total cost
- **TR** — total revenue
- **FC, VC** — fixed cost, variable cost
- **AFC, AVC, ATC** — average fixed, average variable, average total cost
- **LRATC** — long-run average total cost
- **CS, PS** — consumer surplus, producer surplus
- **E** — elasticity
- **SR, LR** — short run, long run
- **Δ** — change in

Chapter 1 — The Principles and Practice of Economics

§1.1 The Scope of Economics

Economics — the study of how agents choose to allocate scarce resources.

Scarce resources — resources of limited supply; using them one way forecloses another.

Economic agent — an individual or group that makes choices.

Microeconomics — individuals, households, firms, and single markets.

Macroeconomics — the economy as a whole — growth, unemployment, inflation.

Model — a simplified description of reality used to make predictions.

§1.2 Three Principles of Economics

- **Optimization** — choosing the best feasible option.
- **Equilibrium** — a situation in which nobody would benefit by changing their own behavior.
- **Empiricism** — analysis that uses data to test theories and identify causes.

§1.3 The First Principle: Optimization

Trade-off — more of one thing means less of another.

Opportunity cost — the value of the best alternative given up.

Budget constraint — the choices that exactly exhaust available resources.

Optimization is an approximation of actual behavior, not a claim that choices are perfect.

§1.4 The Second Principle: Equilibrium

In equilibrium: quantity demanded equals quantity supplied; sellers' costs are covered by the price; buyers value the good at least at the price.

In equilibrium neither side gains by changing its own behavior.

Free riding — enjoying a benefit that cannot be withheld, without contributing to its cost.

§1.5 The Third Principle: Empiricism

Positive analysis — what is; testable against evidence.

Normative analysis — what ought to be; rests on a value judgment.

Optimization is used both positively and normatively.

Correlation is not causation.

Chapter 3 — Optimization

§3.1 Choosing the Best Feasible Option

Feasible option — affordable and achievable given the constraints.

Net benefit — total benefit minus total cost.

$$\text{Net benefit} = \text{total benefit} - \text{total cost}$$

The optimum is the feasible option with the highest net benefit. Feasibility is checked before ranking.

§3.2 Optimization Using Total Value (in Levels)

Compute the total value or total cost of each whole option and compare directly.

Time is costed at the value of its next-best use, usually the wage.

§3.3 Optimization Using Marginal Analysis (in Differences)

Marginal benefit — the additional benefit from one more unit.

Marginal cost — the additional cost from one more unit.

$$\text{Expand while } MB \geq MC. \quad \text{Optimum where } MB = MC$$

Optimization in levels and optimization in differences give identical answers.

Marginal versus average — marginal is the value of one more unit; average is value per unit across all units.

Graphs and Tables

- Total-benefit or net-benefit column: difference consecutive rows for marginal values.
- Marginal benefit and marginal cost table: the last row where MB ≥ MC.
- Marginal benefit bars against a flat marginal cost: the crossing point.

Chapter 4 — Demand, Supply, and Equilibrium

§4.1 Markets

Market — a group of agents trading a good or service, plus the rules and arrangements for trading.

Perfectly competitive market — sellers trade identical goods; no individual buyer or seller is large enough to

affect the market price; entry and exit are free.

Price taker — accepts the market price as given; faces a flat demand curve.

Price maker — can influence the price; faces a downward-sloping demand curve.

Ceteris paribus — holding all else equal.

§4.2 How Buyers Behave

Demand schedule — a table of quantity demanded at each price.

Demand curve — the plot of that schedule; downward sloping.

Law of demand — quantity demanded rises when price falls, ceteris paribus.

Willingness to pay — the most a buyer will pay for a unit.

Diminishing marginal benefit — each additional unit is worth less than the one before.

Market demand is the horizontal sum of individual demand curves: at a fixed price, add the quantities.

A change in the good's own price is a movement along the curve. Every other change shifts it.

- **Five demand shifters** — tastes and preferences; income and wealth; price and availability of related goods; number and scale of buyers; beliefs about the future.

Substitutes: price of one rises, demand for the other rises. Complements: price of one rises, demand for the other falls.

Normal good: income rises, demand rises. Inferior good: income rises, demand falls.

§4.3 How Sellers Behave

Supply schedule — a table of quantity supplied at each price.

Supply curve — the plot of that schedule; upward sloping.

Law of supply — quantity supplied rises when price rises, ceteris paribus.

Willingness to accept — the least a seller will accept for a unit; equal to the marginal cost of that unit.

Market supply is the horizontal sum of individual supply curves.

- **Four supply shifters** — prices of inputs; technology; number and scale of sellers; beliefs about the future.

A change in the good's own price is a movement along the curve. Every other change shifts it.

§4.4 Supply and Demand in Equilibrium

Competitive equilibrium — the price at which quantity demanded equals quantity supplied.

Excess demand (shortage) — quantity demanded exceeds quantity supplied; occurs below the equilibrium price.

Excess supply (surplus) — quantity supplied exceeds quantity demanded; occurs above the equilibrium price.

One curve shifts: equilibrium price and quantity are both determined.

Both curves shift in opposite directions: one of the two is determined, the other is ambiguous.

§4.5 When Government Dictates the Price

Price ceiling — a legal maximum price; binding below equilibrium; creates a shortage.

Price floor — a legal minimum price; binding above equilibrium; creates a surplus.

Non-price rationing — allocation by queues, first-come-first-served, favoritism, or lottery when price cannot clear the market.

Graphs and Tables

- Demand and supply schedules: equilibrium, and horizontal summation into market curves.
- Shift diagram: which curve moved, which way, and the effect on equilibrium price and quantity.
- Binding ceiling or floor: quantity demanded and quantity supplied at the controlled price, and the gap between them.

Chapter 5 — Consumers and Incentives

§5.1 The Buyer's Problem

The buyer's problem has three parts: tastes and preferences, prices, and the budget.

Budget set — every bundle the buyer can afford.

Budget constraint — the bundles that exhaust the budget; the boundary of the budget set. The set is larger than the constraint.

$P_1Q_1 + P_2Q_2 = \text{income}$ **Slope (good 1 on the horizontal axis) $= -P_1 / P_2$**

An income change shifts the constraint in parallel. A change in one price pivots it and changes the slope.

§5.2 Putting It All Together

An optimizing buyer equates the marginal benefit of the last dollar across every good purchased.

$MB_1 / P_1 = MB_2 / P_2 = \ldots$ for every good bought

Marginal utility is the same idea as marginal benefit.

If marginal benefit per dollar is higher on one good, spending shifts toward it.

§5.3 From the Buyer's Problem to the Demand Curve

The demand curve is traced by solving the buyer's problem at each price and recording the optimal quantity.

A price change pivots the budget constraint and moves the optimal bundle.

Market demand is the horizontal sum across buyers.

§5.4 Consumer Surplus

Consumer surplus — willingness to pay minus price paid, summed over the units bought; the area below the demand curve and above the market price.

CS on a straight-line demand: ½ × base × height, base = quantity, height = choke price − market price

Because marginal benefit diminishes, each additional unit adds less surplus than the one before.

§5.5 Demand Elasticities

Price elasticity of demand — the responsiveness of quantity demanded to a change in price.

$E = \%\Delta Q / \%\Delta P$ **Midpoint method: each change divided by the average of the old and new values**

$|E| > 1$ elastic · $|E| < 1$ inelastic · $|E| = 1$ unit elastic · vertical curve perfectly inelastic · horizontal curve perfectly elastic.

Total revenue: elastic demand, a price cut raises revenue; inelastic demand, a price rise raises revenue; unit elastic, total spending is unchanged.

Demand is more elastic with more substitutes, a larger share of the budget, and more time to adjust.

Cross-price elasticity $= \%\Delta Q$ of A / $\%\Delta P$ of B

positive → substitutes · negative → complements · zero → unrelated

Income elasticity $= \%\Delta Q / \%\Delta$income

negative → inferior · between 0 and 1 → normal necessity · above 1 → normal luxury

Graphs and Tables

- Budget constraint before and after a change: slope, position, or both.
- Demand curve with a price line: consumer surplus as an area, and its change when price moves.
- Total-benefit table across goods: the affordable bundle with the highest total benefit.
- Elasticity table: the elasticity multiplied by the percentage change, and the sign.

Chapter 6 — Sellers and Incentives

§6.1 Sellers in a Perfectly Competitive Market

Conditions: sellers produce identical goods; no buyer or seller is large enough to influence the market price; there is free entry and exit.

Free entry — entry unobstructed by legal or technical barriers.

Price taker — a seller too small to affect the market price.

An individual firm faces a horizontal demand curve at the market price. The industry faces a downward-sloping demand curve.

§6.2 The Seller's Problem

The seller's problem has three parts: production, costs, and revenues.

Production function — the relationship between inputs and the output they produce.

Marginal product — the additional output from one more unit of an input.

Specialization — the range over which marginal product is rising.

Diminishing returns — the range over which marginal product is falling but still positive.

Beyond that range marginal product is negative and total output falls.

Fixed cost — does not vary with output. Variable cost — varies with output.

Sunk cost — a cost already committed that can never be recovered; irrelevant to current decisions. Fixed costs are sunk in the short run.

Explicit cost — a direct monetary payment. Implicit cost — the opportunity cost of a resource already owned.

Accounting profit = TR − explicit costs **Economic profit = TR − explicit costs − implicit costs**

Economic profit is never larger than accounting profit. Zero economic profit means positive accounting profit.

AFC = FC/Q · AVC = VC/Q ·
ATC = TC/Q = AFC + AVC
MC = ΔTC/ΔQ
Profit = (P − ATC) × Q

MC cuts AVC and ATC at their minimum points, from below. Fixed cost is the gap between ATC and AVC multiplied by quantity.

Economies of scale — LRATC falls as output rises.

Constant returns to scale — LRATC is flat.

Diseconomies of scale — LRATC rises.

Specialization is one source of economies of scale.

§6.3 From the Seller's Problem to the Supply Curve

Profit-maximizing output: P = MC
Firm's short-run supply curve: MC at every price at or above minimum AVC

Price elasticity of supply — %ΔQ supplied / %ΔP; positive, because a higher price raises quantity supplied.

A supply curve is unit-elastic when it is a straight line through the origin, whatever its slope.

Supply is more elastic with more time to adjust and more spare capacity.

§6.4 Producer Surplus

Producer surplus — price received minus willingness to accept, summed over the units sold; the area above the supply curve and below the market price.

Producer surplus is not profit: it subtracts variable costs only, while profit also subtracts fixed costs.

PS on a straight-line supply: ½ × base × height, base = quantity sold, height = price − supply

intercept

§6.5 From the Short Run to the Long Run

Short run: at least one input is fixed, and fixed costs are paid whether or not the firm produces.

Long run: every input is adjustable and the firm may leave the industry.

Shut down (short run) if $P <$ minimum AVC

Exit (long run) if $P <$ minimum ATC

A firm that shuts down still pays its fixed cost. A firm that exits does not.

§6.6 Long-Run Competitive Equilibrium

Economic profits attract entry, shifting market supply right and lowering the price.

Losses cause exit, shifting market supply left and raising the price.

Long-run equilibrium: $P =$ minimum ATC and economic profit $= 0$

Graphs and Tables

- Cost-curve diagram with MC, AVC, ATC and MR: profit-maximizing quantity, shutdown point, break-even point, and fixed cost, profit or loss as an area.
- Total-output table: marginal product, and the boundaries between specialization, diminishing returns, and negative returns.
- Supply schedules for individual firms: horizontal summation, and willingness to accept for a given unit.
- Supply curve with a price line: producer surplus as an area, and its change when price moves.
- Industry diagram with entry or exit: which curve moves, which way, and where the process stops.
- LRATC curve: economies of scale, constant returns to scale, diseconomies of scale.

Key Formulas

- Net benefit = total benefit − total cost
- Optimum at the margin: expand while MB ≥ MC; optimum where MB = MC
- Opportunity cost of time = hours × value per hour
- Budget constraint: $P_1Q_1 + P_2Q_2 =$ income; slope $= -P_1/P_2$
- Equal-marginal principle: $MB_1/P_1 = MB_2/P_2$ for every good bought
- Consumer surplus = willingness to pay − price paid
- Producer surplus = price − willingness to accept
- Triangle = ½ × base × height · rectangle = base × height
- Price elasticity = $\%\Delta Q / \%\Delta P$, midpoint base = average of old and new
- Cross-price elasticity = $\%\Delta Q_A / \%\Delta P_B$ · income elasticity = $\%\Delta Q / \%\Delta$income
- Market demand or supply = horizontal sum of individual curves
- Accounting profit = TR − explicit · economic profit = TR − explicit − implicit
- $MC = \Delta TC/\Delta Q$ · $AVC = VC/Q$ · $ATC = TC/Q$ · $AFC = FC/Q$
- Fixed cost = (ATC − AVC) × Q
- Profit = (P − ATC) × Q
- Marginal product = Δoutput / Δinput
- Firm supply: P = MC at prices at or above minimum AVC
- Shut down if P < min AVC · exit if P < min ATC
- Long-run equilibrium: P = min ATC, economic profit = 0